**SDC HACKATHON 2026**

# Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | sticks and stones |
| **Project / Product Name** | Resolve |
| **Final Live Deployment URL** | https://resolve-sdc-hackathon-src.vercel.app |
| **Codebase Repository URL** | https://github.com/ejhchesteree7-code/Resolve-SDC-HACKATHON- |
| **Name (Leader)** | EE JUN HAO |
| **Email & phone (Leader)** | ejhchesteree7@gmail.com (011 – 1615 1223) |

## 2. Project Description

***[Maximum 250 words]***

Owner-run Malaysian SMEs lose customers through reviews nobody handles. Reviews are scattered, replies are late, nobody is sure who may offer compensation, and the same complaint repeats unnoticed. In our Phase 1 test, replying by hand took 72.5 seconds per review and seven manual steps.

Resolve turns every review into a tracked case with an owner. A guest scans the outlet's QR code or taps an NFC card and leaves a review. Resolve checks it for duplicates and spam, uses GenAI to classify it (Bahasa Melayu, English or Manglish) and draft a reply with a one-line reason, then routes it with plain rules: Staff for praise, a manager for complaints, and the Owner for anything about safety, harassment or money. Nothing is posted and no voucher offered until the right role approves it, and every step is written to an activity log the database itself locks against edits.

For the final phase, we turned it into a product a real business can sign up for: logins for each team member, one-page registration with a QR code per outlet, a Team page to remove access the day someone leaves, overdue alerts, and an Impact page measuring handling time, approval coverage and routing accuracy from the system's own records. Built for walk-in SMEs with small teams and no customer-service department, ready-made for cafés, salons, retail shops, car workshops and pet care on the same approval rules.

Resolve supports SDG 8 by giving small teams clear roles and owners their time back.

# 3. Upgraded Product & Technical Execution

## 3.1 Final System Architecture

*Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.*

Resolve is one TypeScript monorepo deployed on Vercel: a React + Vite web app, an Express API running as a Vercel serverless function, and a Supabase Postgres database (row-level security on, no public policies). The diagram below shows the flow.

**1. Review in.** Every review enters through one endpoint, POST /api/events/review-received, whether it comes from the guest page opened by an outlet's QR code or NFC card, or in production from Orbyte, Google Business Profile or WhatsApp.

**2. Duplicate check (R0).** Before any AI call, Resolve flags identical text within 24 hours, the same reviewer within 30 minutes, or a burst of five reviews in five minutes. Flagged cases go to at least a Manager and never get a voucher.

**3. GenAI step.** The API calls an OpenAI-compatible model (KrackedDevs credits) with the business type's category guide. The model must return strict JSON (language, sentiment, category, severity, a draft reply and a one-line reason), validated before use, with a 12-second timeout. If the call fails, is slow or returns invalid JSON, Basic mode (keyword rules and reply templates per business type) takes over, so the workflow never depends on the AI being available.

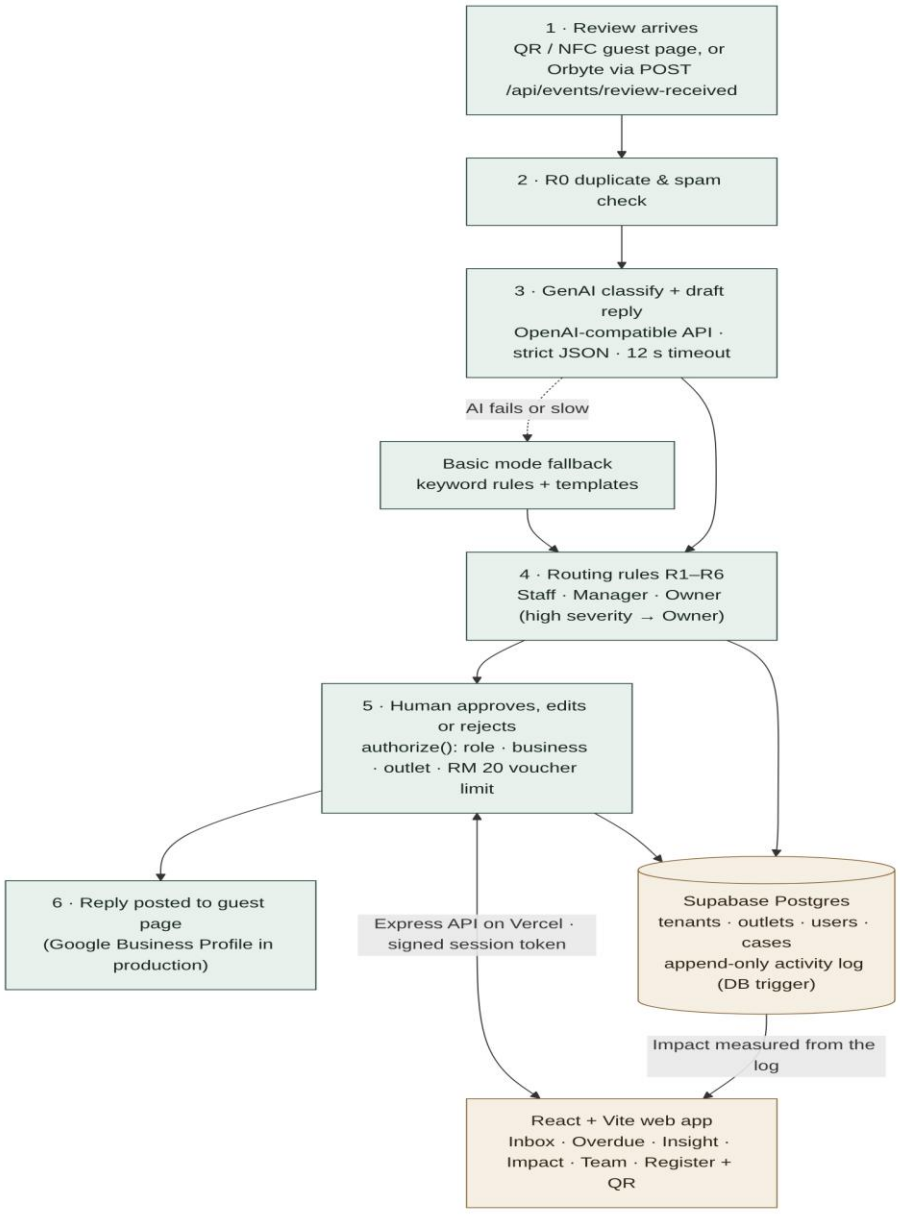
**4. Routing rules R1–R6.** Plain, explainable rules choose the role: high severity (safety, harassment, money disputes) always goes to the Owner; vouchers above the business's limit (default RM 20) go to the Owner; negative reviews go to a Manager; simple praise goes to Staff.

**5. Human approval.** Every action (approve, edit, reject, voucher, guest follow-up) passes one server-side authorize() check of the user's role, business and outlet inside the same database transaction. Users log in with salted scrypt password hashes and HMAC-signed 12-hour session tokens; the Owner can remove a person's access instantly.

**6. Records.** Postgres tables hold tenants (businesses), outlets, users, cases and the activity log. The log is append-only: a database trigger rejects every UPDATE and DELETE, so even the app cannot rewrite history.

**7. Measurement.** The web app times each approval and writes the handling time into the log. The Impact page reads the log to report median handling time, approval coverage and exceptions, and reruns a 55-review labelled test set through the same classifier and rules.

Five business types (café, hair salon, retail shop, car workshop, pet care) share one set of categories and rules; only the words, sample reviews and reply templates change.


1 · Review arrives
QR / NFC guest page, or
Orbyte via POST
/api/events/review-received
2 · R0 duplicate & spam
check
3 · GenAI classify + draft
reply
OpenAI-compatible API ·
strict JSON · 12 s timeout
AI fails or slow
Basic mode fallback
keyword rules + templates
4 · Routing rules R1–R6
Staff · Manager · Owner
(high severity → Owner)
5 · Human approves, edits
or rejects
authorize(): role · business
· outlet · RM 20 voucher
limit
6 · Reply posted to guest
page
(Google Business Profile in
production)
Express API on Vercel ·
signed session token
Supabase Postgres
tenants · outlets · users ·
cases
append-only activity log
(DB trigger)
Impact measured from the
log
React + Vite web app
Inbox · Overdue · Insight ·
Impact · Team · Register +
QR

### 3.2 Implementation & Hackathon Deliverables

*Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.*

***[Maximum 300 words]***

Phase 1 was a working workflow with a "Viewing as" role switcher, one café and a manual timing test. In the finalist round we built what a real owner would need first.

**Built, integrated and deployed (live on Vercel with Supabase):**

1. **Real logins:** salted scrypt passwords (one-way hashed, never plain text) and signed session tokens replace the role switcher, so nobody can pick their own role.
2. **Business registration:** one page creates the business, first outlet, owner and team logins, and a downloadable QR code for the outlet's review page.
3. **Team page:** the owner adds people, resets passwords and removes access; everyone can change their own password.
4. **Five SME types on one set of rules:** café, hair salon, retail shop, car workshop and pet care. Only the words, sample reviews and reply templates change.
5. **Duplicate and spam check (rule R0):** repeated text, repeat reviewers and bursts are flagged for a Manager and never get a voucher.
6. **Overdue alerts:** urgent cases after 2 hours, others after 24 hours.
7. **Impact page:** handling time, approval coverage and routing accuracy, measured from the activity log and a 55-review labelled test set.
8. **Live GenAI** through an OpenAI-compatible API, with Basic mode as a fallback.

**Problem-solving steps:**

- A first-round mentor compared us to Trustpilot. Our answer: Trustpilot displays reviews, while Resolve handles them, with roles, owner escalation, voucher limits and a locked log, built for Malaysian SMEs.
- Supporting more business types risked rewriting the safety rules for each one, so we mapped every industry onto shared categories; a burnt scalp or an injured pet is routed exactly like a food-safety complaint.
- Phase 1 timings came from a one-off manual test, so we now time each approval inside the app and write it into the log.

## 4. Environmental & SDG Impact Verification

*Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.*

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

| Metric Category | Your Response |
|---|---|
| **Target SDG Metric Achievement**<br>*Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers*<br>*e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."* | SDG 8 (Decent Work and Economic Growth), against our four Phase 1 KPIs:<br>• **Handling time** (target: at least 60% faster than the 72.5 s manual baseline). Phase 1 timed test: 27 s per review, 62.7% faster. Phase 2, measured inside the app: median **[12.5]** s per review, **[82.8]**% faster, across **[5]** logged approvals.<br>• **Correct routing** (target: at least 90%). Achieved 100% (55 of 55 labelled reviews across all five business types), with 98.2% correct categories and 15 of 15 urgent cases reaching the Owner.<br>• **Replies and vouchers with a logged human approval** (target: 100%). Achieved 100% (**[18 of 18]** actions).<br>• **First-time users resolving a complaint unaided** (target: 80%). [Not yet measured; planned with real café teams in our 30-day pilot.]<br><br>**Manual steps per review:** 7 by hand, 2 with Resolve (71% fewer). |
| **Verification Evidence Method**<br>*How was this data verified?* | Database transaction counters and the append-only activity log. Each approval records its handling time in milliseconds |

| Metric Category | Your Response |
| --- | --- |
| *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.* | (timed from opening a case to clicking Approve), and the Impact page computes the median from those log entries. Approval coverage checks that every "reply posted" or "voucher issued" entry has a matching "approved" entry by a Staff, Manager or Owner on the same case. Routing accuracy comes from a 55-review labelled synthetic test set (English, Malay and Manglish, five business types), labelled by the team before checking Resolve's output and re-run through the same code on every refresh. A database trigger blocks edits and deletes on the log, so the figures cannot be changed after the fact. All data is synthetic. |
| **Socio-Economic / Civic Value**<br>*How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?* | Small owner-run businesses are a large share of Malaysia's employers, yet most have no system for customer feedback. Resolve gives each team member a clear, limited role, protects staff from being blamed for decisions above their level (vouchers above RM 20 and safety complaints go to the Owner), and saves owners time each week on replies. Because a business is configured rather than coded, the same workflow serves cafés, salons, shops, workshops and pet care businesses, and could be offered through partners such as Orbyte, SME associations or Rotary business networks to many SMEs at once.<br>PDPA: Guests see a consent notice before submitting and may use an alias. We collect only a first name, rating and review; the AI never sees names. Role-based access is enforced on the server (Staff and Managers see only their outlet; only the Owner manages the team). Reviews are anonymised after 12 months. The prototype uses synthetic data. |
| **Deployment & Commercial Feasibility**<br>*e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.* | Resolve is deployed as a web app with a documented event interface: one event in (review received) and one event out (reply posted after human approval). That lets it run as a plug-in module inside an existing platform such as Orbyte, with Google Business Profile and WhatsApp Business connectors replacing the simulated steps. The planned business model is RM 49 per outlet per month, plus an optional RM 30 starter kit (QR counter stand and NFC card).The AI step is one short call per review, and Basic mode keeps the workflow running at no AI cost if the API is unavailable. |

### 4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

*Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.*

*This proof is graded across three categories at once:* ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

**Formulas (as implemented)**

```
handling reduction (%) = (1 - median handling time / 72.5 s) x 100
weekly minutes saved   = (72.5 s - median) x 30 reviews per week / 60
approval coverage (%)  = actions with a matching human "approved" entry
                         / all reply and voucher actions x 100
routing accuracy (%)   = test reviews routed to the expected role / 55 x 100
```

**Code: src/artifacts/api-server/src/lib/resolve-impact.ts**

```
const timings = activity
  .filter(row => row.action === "approved" && HUMAN_ROLES.has(row.role))
  .map(row => handledPattern.exec(row.details)?.[1])   // "handled in X s"
  .filter((value): value is string => value !== undefined)
  .map(Number);
const mid = median(timings);
reductionPct = round1((1 - mid / PHASE1.baselineSeconds) * 100);   // 72.5 s
weeklyMinutesSaved = round1(((PHASE1.baselineSeconds - mid) * REVIEWS_PER_WEEK) / 60);


const approvedCases = new Set(activity
  .filter(row => row.action === "approved" && HUMAN_ROLES.has(row.role))
  .map(row => row.reviewId));
const posted = activity.filter(row =>
  row.action === "reply_posted" || row.action === "voucher_issued");
const postedWithApproval = posted.filter(row => approvedCases.has(row.reviewId)).length;


for (const item of evalSet) {               // 55 labelled reviews, 5 business types
  const result = await classifyNewReview({ text: item.text, rating: item.rating,
    businessType: item.businessType }, { aiDisabled: true });
  const routed = routeNewReview(result, result.category === "unclear");
  if (routed.assignedRole === (item.urgent ? "Owner" : "Manager")) routing++;
}
```

Worked example: with a measured median of 12.5 s, the reduction is (1 − 12.5 ÷ 72.5) × 100 = 82.8%, and the time saved is (72.5 − 12.5) × 30 ÷ 60 = 30 minutes per outlet per week.

Sources:

1) Bernama. (2026, July 31). Govt shifts focus to scaling up MSMEs after resilient 2025 [News article]. https://www.bernama.com/en/news.php?id=2588561

2) BrightLocal. (2026, February 11). Local consumer review survey 2026. https://www.brightlocal.com/research/consumer-review-survey/

3) Human Resources Online. (n.d.). Malaysia's MSMEs outgrew the national economy in 2025, accounting for nearly 40% of GDP.

https://www.humanresourcesonline.net/malaysia-s-msmes-outgrew-the-national-economy-in-2025 accounting-for-nearly-40-of-gdp

4) United Nations. (n.d.). Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

https://sdgs.un.org/goals/goal8

## 5. Market Viability & Future Scalability Plan

*SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?*

***[Maximum 250 words]***

Resolve scales by configuration, not rewrites. A business is a row of settings (type, outlets, team, voucher limit), so onboarding takes one registration page and a QR code, and a new industry is one profile of words and templates on the same routing rules.

With funding or partners, we would scale in four steps:

1. **Real channels.** Apply for Google Business Profile API access so approved replies post directly to Google, and add WhatsApp Business Platform follow-ups, plugging into events Resolve already exposes: review received and reply posted.
2. **Trust and anti-bot.** Extend our duplicate and spam rule (R0) with a bot challenge on the review page, per-device rate limits and rotating outlet QR tokens, so only customers at the outlet can review. Suspected bot reviews go to a Manager and never earn a voucher.
3. **Orbyte integration.** Use Orbyte's single sign-on and tenant data instead of our own accounts, and offer Resolve as a module for its SME customers.
4. **Pilot and volume.** Run a 30-day pilot with 10 SMEs across three industries via Rotary and XMUM partners, measuring handling time, approval coverage and repeat complaints against each business's baseline. Move AI calls to a queue with batching and caching, keep Basic mode as fallback, and add per-tenant API keys and signed webhooks.

## 6. Final Submission Checklist

*Tick all boxes that apply before submitting.*

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

## 7. IP & Originality Declaration

### 7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☑ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

- The idea of collecting reviews through an NFC card or QR code, and per-staff team logins, comes from ReviewHub (PandaPulse), a product owned by team leader Ee Jun Hao before the hackathon. No ReviewHub code was copied; all Resolve code was written for this hackathon.

### 7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name Type (code/dataset/model) License

| Name | Type (code/dataset/model) | License |
|---|---|---|
| React, React DOM | code | MIT |
| Vite | code | MIT |
| Express | code | MIT |
| Drizzle ORM | code | Apache-2.0 |
| node-postgres (pg) | code | MIT |
| Zod | code | MIT |
| TanStack Query | code | MIT |
| Tailwind CSS | code | MIT |
| wouter | code | Unlicense |
| lucide-react | code | ISC |
| qrcode | code | MIT |
| OpenAI Node SDK | code | Apache-2.0 |
| esbuild, pino | code | MIT |
| Supabase (hosted PostgreSQL) | service / database | PostgreSQL License |
| gpt-5.4-mini via KrackedDevs API. | model (API) | Provider terms of use |
| Synthetic reviews and 55-review test set | dataset | Created by the team |

### 7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

## 8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

[Signature] 

[Date] 31/10/2026